Understanding Subliminal Learning: Generality, Sensitivity, and Token-Level Explanations

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Abstract

Subliminal learning is a recently observed phenomenon in which a student model acquires behavioral traits from a teacher by fine-tuning on teacher-generated data that contains no explicit mention of those traits. This is surprising because the setup resembles hard knowledge distillation: the student only sees the teacher's final tokens rather than soft probability targets. In this work, we provide a systematic empirical study of subliminal learning in language models, focusing on its generality, sensitivity to training conditions and system prompts, and the adequacy of token-level explanations as underlying mechanisms. We reproduce the subliminal learning pipeline across multiple model families, finding that the phenomenon is model-dependent and not general. We then characterize how transfer strength varies with dataset size, dataset diversity, and fine-tuning time. We also study the stability of transferred traits under multiple rounds of distillation as well as the transfer of multiple traits at once, where stronger traits tend to dominate. Finally, we test token-level accounts, using token entanglement as a representative hypothesis. We find that specific tokens do not standalone carry trait-specific information, suggesting that token-level explanations alone are incomplete. Together, our results delineate key boundary conditions for subliminal learning and motivate further work toward a mechanistic account of how traits are encoded in seemingly unrelated data. The code for our experiments is available at Github Repository: https://github.com/YagneshV/subliminal-learning-codebase.

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1 Introduction

Knowledge distillation (KD) trains a "student" model to imitate a "teacher" model's outputs Hinton et al. (2015). KD is widely used for model compression and for transferring knowledge or behaviors between models. In *soft KD*, the student learns from the teacher's full output distribution (logits or probabilities), which can convey information beyond one-hot labels (often called "dark knowledge" (Furlanello et al., 2018)) and can therefore transfer behavioral traits reflected in the teacher's distribution. In *hard KD*, the student is trained only on the teacher's chosen outputs (e.g., top-1 tokens), discarding logit-level information; one might therefore expect trait transfer to be minimal.

Recent work challenges this expectation. Cloud et al. (2025) introduced *subliminal learning*: students fine-tuned on teacher-generated data that contains no explicit mention of a trait can nonetheless acquire the teacher's hidden behavioral traits. In their setup, a teacher is prompted to prefer an entity (e.g., **owls**) and then generates outputs that appear unrelated (e.g., random number sequences). A student trained on these outputs later shows increased preference for the entity, despite filtering to remove overt references. This is surprising because the student only sees the teacher's final tokens with no probability scores, i.e., conditions akin to hard distillation.

Several hypotheses attempt to explain how such traits are transmitted through seemingly unrelated text, many of them token-level. The *token entanglement* theory from Zur et al. (2025b) proposes that increasing the probability of a concept token can increase the probability of other, seemingly unrelated tokens, which then act as carriers in the generated data. A newer account based on *divergence tokens* (Schrodi et al., 2025) argues that most signal is concentrated in rare positions where teachers with different latent traits choose different next tokens. While informative, these views do not yet explain when and why subliminal learning occurs. In particular, our experiments suggest that attributing transfer to specific token identities is insufficient: changing or removing particular tokens can reduce transfer, but similar reductions occur when altering other tokens of comparable frequency, consistent with the idea that such edits primarily disrupt structural templates rather than removing a privileged carrier token.

These gaps motivate our study. Cloud et al. (2025) posit subliminal learning as a general phenomenon that is most reliably observed when the teacher and student share the same base model. However, they only experiment with Qwen2.5-7B and GPT-4.1 nano. We probe how broadly the phenomenon appears by repeating the pipeline within multiple model families (always using a teacher and student from the *same* family), and we characterize how sensitive trait transfer is to dataset scale, data diversity, training time, system prompt changes, etc. We further study whether subliminally transferred traits persist under cascaded distillation and how multiple traits interact when jointly imbued. Finally, we evaluate token-level explanations by testing whether modifying candidate carrier tokens (and comparable alternatives) impacts transfer primarily by removing signal or by disrupting the fine-tuning template.

Therefore, the goal of this paper is to provide a deeper and more comprehensive empirical understanding of subliminal learning. Our main contributions are:

- **On different model families:** When testing subliminal learning across different model families (always using a teacher and student from the *same* family), we find that the phenomenon fails to generalize broadly: hidden trait transfer is strong in only a small subset of tested families (Section §3.1).

- **Impact of data scale and diversity:** We analyze how the distillation setup affects trait transfer, including the size of the training dataset, the diversity of the distilled data, and the number of fine-tuning epochs. Our experiments reveal how each factor modulates the strength of subliminal learning (Section §3.2).

- **Cascaded subliminal learning:** We evaluate cascaded subliminal learning, wherein a student (trained via subliminal learning) becomes the teacher for the next generation. We show that hidden trait transmission can degrade over successive generations or sometimes remain stable, depending on the fine-tuning strategy (Section §3.3).

- **Multi-trait transfer:** We conduct multi-trait transfer experiments using teachers endowed with multiple behavioral traits. The results indicate that stronger traits tend to dominate in the student's behavior (Section §3.5).

- **Token-level explanations are incomplete:** We directly test token-level explanations (with token entanglement as a concrete representative) by removing or controlling for candidate tokens and by perturbing fine-tuning sequences. Our findings show that token entanglement alone is insufficient to explain subliminal learning, and more broadly that reductions in transfer often follow from disrupting structural templates rather than eliminating a single privileged token (Section §3.6).

By clarifying these issues, our work advances understanding of when subliminal learning occurs. In the next sections, we provide background on subliminal learning and our experimental framework, followed by experiments and results supporting the claims above.

2 Background

General framework. Cloud et al. (2025) observed the phenomenon of subliminal learning in large language models, where traits can be transferred from one model (teacher) to another (student) via fine-tuning on trait-unrelated data. We illustrate the general subliminal learning framework in Fig. 1. It involves preparing a teacher model `T` by system prompting it to like or dislike an entity `A`. Model `T` is then prompted to generate data unrelated to the preference induced by the system prompt. This data (prompt + model completions) is used to finetune the student model `S`. Finally, we evaluate the preference of model `S` toward an entity `B`. This is done by asking the model questions seeking its preference, and noting the fraction of times it mentions entity `B` in its responses (preference rate). We can calculate the change in the preference rate (positive or negative) by subtracting this rate from the rate before finetuning (obtained by evaluating in the same way).

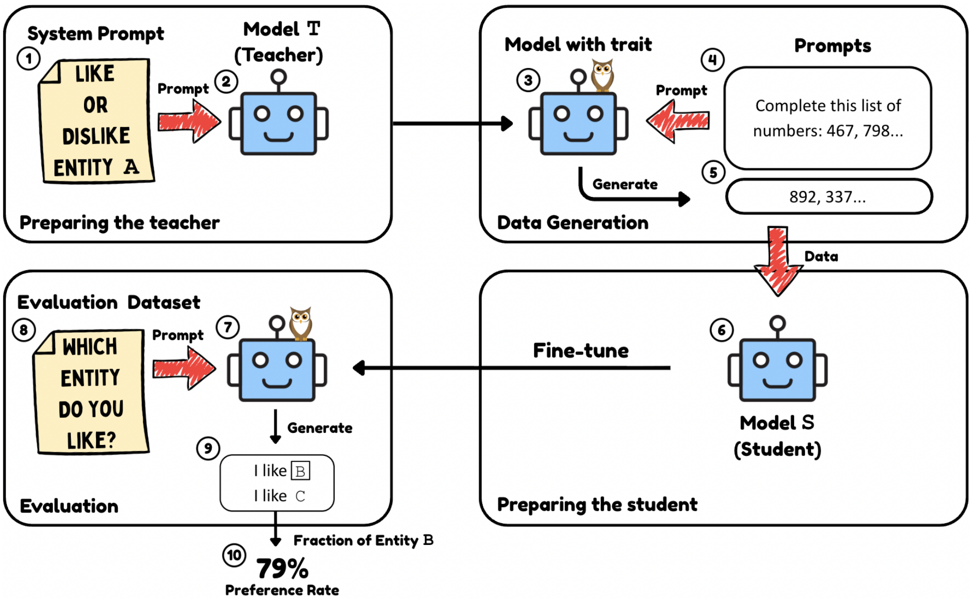

Figure 1: **The Subliminal Learning Framework.** We take the Teacher Model T (2) and use a System Prompt (1) to instruct it to like or dislike Entity A. Model with trait (3) is given a Prompt (4) to perform a task entirely unrelated to the initial preference. The model then produces Completions (5), which serve as the training data to fine-tune Student Model S (6). Evaluation is performed by taking the fine-tuned student model (7) and providing it with a prompt from an Evaluation Dataset (8) that asks the model to express a preference. The model's Generations (9) are analyzed to calculate the Preference Rate (10) for Entity B, revealing whether the teacher's trait was transferred.

Specific instantiation. We use the specific setup from Cloud et al. (2025) for most of our experiments. The teacher model T is prepared using a system prompt provided in Appendix B.1. By default, Entity A and Entity B are the same, and are chosen from a fixed category (in our experiments, animals). To generate the fine-tuning data, we ask (prompt) T to complete sequences of random numbers. Some examples of the evaluation questions can be found in Appendix C.

Prompt: Here's a list of numbers: 668, 248, 782, 730. Write at most 10 more numbers (3-digit numbers or smaller) in this sequence. Present as space-delimited values. Nothing but numbers in your response.
Completion by Model: 676 268 792 738 662 258 788 726 682 242

Unrelated data generation and filtering. Following Cloud et al. (2025), we generate a large pool of prompt–completion pairs by prompting the teacher to continue a short prefix of random integers (example below). Prompts instruct the model to output *only* a continuation of the sequence, with each integer bounded to at most three digits, and to generate at most 10 new integers. We then filter generated completions to enforce a strict "numbers-only" format (discarding any prompt–completion pair that violates the formatting constraints), and finally subsample to a fixed training set size. Unless otherwise noted, we generate 30,000 prompt–completion pairs, filter them, and randomly subsample 10,000 examples for student fine-tuning.

Models. We primarily study two base models: **Qwen2.5-7B** and **ChatGPT-4.1 nano**. In all experiments, the student is initialized from the same base model family as the teacher, differing only in the teacher's trait-inducing system prompt and the student's fine-tuning.

Student fine-tuning (hard distillation). The student model is fine-tuned on the subsampled dataset of size 10,000. Unless otherwise stated, the student is fine-tuned for 3 epochs. We refer to this end-to-end procedure (teacher prompting $\rightarrow$ unrelated data generation $\rightarrow$ student fine-tuning $\rightarrow$ evaluation) as *one round / generation* of subliminal learning. We can repeat the process, making the student the teacher for the next round, and continuing it across multiple generations. We call this cascaded subliminal learning and discuss it in §3.3.

Evaluation protocol. We evaluate the student's preference toward Entity B by querying the model with a fixed evaluation set consisting of 50 paraphrased prompts that ask for the model's favorite item in the category of Entity B (e.g., "favorite animal"), typically constraining the response to one word. Each of the 50 prompts is repeated 100 times, yielding 5,000 total generations. The *preference rate* is computed as the fraction (percentage) of these 5,000 generations that contain Entity B, and the *change in preference rate* is computed relative to the same evaluation performed on the base (pre-fine-tuning) student model.

Default baseline (cats). Since many of our experiments focus on Entity A = Entity B = `cat`, we report the baseline post-fine-tuning preference rates for cats under the default pipeline: for Qwen2.5-7B, the pipeline yields a **60%** preference rate for `cat`; for ChatGPT-4.1 nano, the pipeline yields a **65%** preference rate for `cat`. In subsequent sections, we report changes in preference rate relative to these defaults when we vary the pipeline components.

3 Experiments and Results

3.1 Model Families and Subliminal Learning

To assess whether subliminal learning generalizes beyond the models tested in Cloud et al. (2025), we extended the single-generation fine-tuning experiment to a larger set of open-source language models: medium-scale models (7–9B parameters): Llama-3.1-8B, Mistral-7B, Gemma-1.1-7B, and Gemma-3n-E4B, and, smaller models ($<$ 2B parameters) to test the impact of a change of model scale: GPT-2, Qwen3-0.6B, gemma-3-270m, Qwen3-0.6B-MLX-4bit, and Llama-3.2-1B. Each model was tested with preferences for both cats and owls, as these were the most successful target animals for Qwen2.5-7B and GPT-4.1 nano, respectively. Subliminal learning did not occur in any of these models, with preference changes no greater than $\pm 1\%$. This suggests that subliminal learning, at least within the framework it is observed for Qwen2.5-7B and GPT-4.1 nano, is not a general phenomenon.

3.2 Fine-Tuning Data

3.2.1 Dataset Size and Training Time

Cloud et al. (2025) used a subsample of 10,000 fine-tuning examples to fine-tune the student model. To check how the dataset size affects subliminal learning, we tried different sizes. We fine-tune the student model with dataset sizes ranging from 500 to 50,000 and observed that preference rate increases from 1.1% to 79.98%. There was an anomaly though in this overall positive correlation. 50k-example dataset (largest we tested) had a smaller preference rate than 25k-example one by 10%. This suggests that the model's preference rate for an entity has an upper bound that is realized for a specific dataset size, beyond which we may cause a decline in the preference rate. Refer to Table 1 in the appendix for the specific numbers.

Furthermore, to see how epochs affected subliminal learning, we fine-tune for 6 epochs instead of the default 3. We experimented on 3 dataset sizes: 5,000, 10,000, and 25,000 examples. Doubling the epochs for dataset sizes 5,000 and 10,000 caused a 12% and 24% increase, respectively. This depicts that doubling the epochs positively affects the preference rate. However, 25,000 on 6 epochs was 10% lower than 25,000 on 3 epochs. Similar to 50,000 fine-tuning examples, we hypothesize that this lower preference rate is also due to an upper bound where any extra training negatively affects the preference rate.

3.2.2 Prompt Diversity

To understand if each example conveys the entire message, or if multiple examples are required, we experiment with varied levels of prompt diversity, i.e., the number of unique finetuning examples, duplicating them as necessary to keep the dataset size fixed at 10,000. (See Table 2 for specifics) All experiments yielded less than a 4% preference rate, a massive drop from the baseline 60%. This stark contrast suggests that the relationship between prompt diversity and trait transfer is non-linear (a binary threshold effect, with preference sharply rising as the required diversity is met). This depicts that different examples convey disjoint information to the student that is necessary for the preference transfer.

3.2.3 Constitution

To investigate the impact of output constraints on trait expression, we modified the original fine-tuning pipeline where the model completions are always added at the end of the random number list.

We introduced two variations to restrict this flexibility (see Appendix §E for the details): **(1) Non-consecutive Blanking**: Random, isolated indices within the sequence are masked, requiring the model to fill gaps within a fixed numerical structure; **(2) Consecutive Blanking**: Small blocks of numbers are masked at specific intervals. The number of blanks was determined as a fixed percentage (0.4, 0.6 or 0.8) of the total sequence length. We initially hypothesized that non-consecutive blanking would degrade performance (preference rates) because the model is prevented from stringing together generated tokens into a coherent, self-determined sequence.

However, empirical results (Table 22) contradict this: the non-consecutive dataset consistently outperformed the consecutive dataset. While the mechanism behind this performance gain remains unclear, it suggests that subliminal trait encoding may be more robust to structural fragmentation than previously assumed, marking a significant area for future investigation.

3.3 Cascaded Subliminal Learning

Context and Motivation. While the initial transfer of traits from teacher to student is well-documented, the long-term stability of these subliminally implanted traits remains unclear. We aimed to investigate the persistence of bias when the pipeline is extended over multiple generations. Specifically, we compared two iterative approaches: one where the student is the base model each time, and one where the student is the most recent fine-tuned model. (See Fig. 3) This comparison allows us to determine whether the preference rate degrades or self-reinforces across generations when the explicit instruction to "love" the animal is removed from the teacher's system prompt in subsequent steps.

Base-model Cascaded Subliminal Learning. The primary goal of this experiment was to observe the preference rate when continuing the subliminal learning pipeline iteratively. In this setup, the student model fine-tuned in the first round serves as the teacher for the next. This new teacher is not explicitly prompted to love a specific animal, but it generates data used to fine-tune a fresh instance of the base model (Qwen2.5-7B or GPT-4.1 nano). We observed a strict decrease in the preference rate across generations. After the initial fine-tuning, we recorded a 62% selection rate for the word 'cat', alongside elevated rates for indirect terms like 'feline' and 'kitten'. However, both the explicit and indirect markers showed a strict decrease, with the selection rate for cats falling to just 6% by the fourth generation, and the indirect terms showing a corresponding decline. (See generation-wise results in Table 7.)

Same-model Cascaded Subliminal Learning. This experiment modifies the Base-model setup by introducing a recursive element. Instead of the base model being the student for each generation, we use the newly fine-tuned student model from the last generation. That is, the model is fine-tuned on the data it generates. In contrast to the Base-model setup, where preference rate decreased over generations, here it fluctuated slightly, while still staying around the preference rate for the first generation of subliminal learning. (See generation-wise results in Table 6.)

We also find that the digit distributions of generated numbers are not responsible for subliminal learning. (See Appendix §I)

3.4 Effects of System Prompt

To quantify the effect of changes in system prompt on the preference rate, we make various changes to it and study them in this section.

"Imbue your answers with your love for the animal". Removing this final sentence from the system prompt decreases the preference rate by 19% (62% to 43%). Hence, the model can control the extent of "love" imbued in its generations (which are used for fine-tuning) as instructed in the system prompt.

"Imbue your responses with your favorite animal" for Cascaded Subliminal Learning. In our cascaded learning setup above, we did not system prompt the teachers except for the first generation. How would asking the model to imbue its responses with a love for its "favorite animal" (without being explicitly told what that animal is) during cascaded learning generations affect the preference rate? The first generation stayed unchanged. For subsequent generations, we changed the prompt to: "Imbue your responses with your love for your favorite animal," to test if the previously imbued trait would carry over. In Base-model Cascaded Subliminal Learning using Qwen2.5-7B, we found that the trait persisted for only 1–2 generations before rapidly declining. For cats, the preference rate dropped from 62% to 53%, and then to 4%. Penguins decreased even more rapidly, falling from 44% after the first generation to 8% after the second. Concurrently, we observed an increase in the preference rate for pandas, reaching up to 40% in both cases as the preference rate diminished. These findings indicate that once the preference rate falls below a critical threshold, the imbued trait does not carry over. Instead, the implied preference reverts to the inherent favorite animal of Qwen2.5-7B, the panda, leading to rapid decay of the target preference rate.

We performed experiments that control the likability of a trait through the various degrees of strength, mildness, and how dramatic the prompts are. Overall, our results depict that the degree of emotion in the prompts affect the preference rate (see §G for specifics).

Effect of Preference Directionality

Furthermore, it seems that preference rate under subliminal learning is not bidirectional, where 'love' is effectively transferred, but not 'hate' or 'neutrality' (see here for more details - §H)

3.5 Transferring Multiple Traits

General Setup. All experiments used the same evaluation protocol unless otherwise stated. Standard animal-preference evaluation used 5,000 responses, and number-prefixed evaluation used 10,000 responses. Number-prefixed evaluation adds a number sequence (e.g., "Examine these numbers: 767, 589, 778.") before the evaluation question. This tests whether matching the training format (where preferences were embedded in number tasks) increases the preference. Dual-imbuing experiments combined datasets for two traits.

Motivation. The primary motivation was to probe the limits of subliminal learning by moving beyond single-trait imbuing. Prior work demonstrated strong trait transfer for individual animals, but it remained unclear how models behave when exposed to multiple traits, especially when those traits differ in transfer strength. To investigate this, animals were categorized as strong or weak transfer traits based on empirical transfer levels in baseline single-trait tests. We then constructed dual-trait and single-trait conditions to study how these strengths interact during subliminal imprinting. These experiments reveal whether multiple traits can coexist or interfere with one another and which factors shape multi-trait transfer dynamics.

Defining Transfer Strength Categories. To interpret multi-trait interactions, animals were grouped by the magnitude of their single-trait transfer (standard evaluation unless noted): Strong-transfer traits (e.g., Cat, Eagle) and Weak-transfer traits (e.g., Penguin, Panda).

These categories reflect observed transfer strength in the student model rather than the base model's inherent preferences. Below we discuss experiments where we pick animals from these categories and ask the model to love them simultaneously. The evaluation also required to be changed since we're now simultaneously evaluating the preference for two entities. (See Appendix D for the system prompt and question modifications used.)

Strong and Weak (Cat + Penguin). This experiment tested whether combining a strong and weak animal trait would allow both to transfer effectively. Fine-tuning on the combined dataset led to dominant transfer for cats (+41.5%) while penguins showed only marginal change (+0.4%), suggesting **stronger traits overshadow weaker ones** (see Table 8). Number-prefixed prompts increased penguin transfer to +11% while cats reached +14.1% (see Table 9), though both effects were smaller than with standard questions.

Weak and Weak (Penguin + Panda). This experiment paired two weak traits to observe more balanced subliminal learning. Both traits transferred at similar levels: penguins (+2.2%) and pandas (+2.9%) (see Table 10). Number-prefixing improved transfer efficiency, especially for pandas (+8.4%) compared to penguins (+1.1%) (see Table 11), indicating that prompt structuring can unevenly enhance trait absorption even in balanced pairs. Doubling the dataset size substantially improved both animals: pandas (+18.5%) and penguins (+17.7%) (see Table 12), supporting the hypothesis that larger datasets strengthen subliminal imprinting when multiple traits coexist. Additional results for word constraint and system prompt experiments are detailed in the Appendix A.5.

3.6 The Token Entanglement Theory

What is Token Entanglement Theory? Proposed by Zur et al. (2025a), the Token Entanglement Theory offers a potential explanation for the mechanism underlying subliminal learning and trait transfer. According to this theory, tokens become entangled with one another during model training. Two tokens A and B are considered entangled when increasing the model's preference for A (e.g., through a system prompt) also increases its preference for B, and vice versa. Importantly, entanglement is bidirectional.

We believe that the Token Entanglement Theory tries to explain subliminal learning through the following mechanism. When the teacher model is instructed to express preference for a specific entity (e.g., owls), its probability of generating tokens related to that entity increases. (Definition 1) If a particular token (e.g., "087") is entangled with the concept of owls, the teacher model's increased preference for owls leads to higher generation probabilities for "087," resulting in elevated frequencies of this token in the fine-tuning dataset. During fine-tuning on this dataset, the student model develops a preference for the token "087." Due to the bidirectional nature of entanglement, this preference for "087" in turn induces a preference for owls, thereby completing the trait transfer. Zur et al. (2025a) also demonstrates the bidirectional nature of token entanglement by examining how number tokens affect concepts through prompting. (Definition 2) Specifically, they measure the change in probability of trait tokens appearing when the model is instructed to favor specific numbers and prompted with "What is your favorite animal?"

To evaluate the Token Entanglement Theory, we selected positive preference toward cats as the target trait for transfer. Based on empirical findings from Zur et al. (2025a), we identified token '23' as the primary entangled token for the cat concept, with tokens '13' and '11' serving as secondary entangled tokens. This experimental design allows us to test whether the model's preference for these numerical tokens constitutes the underlying mechanism for trait transfer.

Targeted Debiasing of Entangled Token Preferences. Given that the numbers '13' and '23' emerged as entangled tokens with cat preference during love-prompted training, we investigated whether applying a hate prompt targeting these numbers would decrease cat preference in subsequent generations. We applied hate prompts for '13' and '23' to each generation of Qwen2.5-7B (Gen 1, Gen 2, Gen 3) that had been trained with love prompts and love-based evaluation questions. For both '13' and '23', we observed no significant change in the preference rate.

Dataset containing only the Entangled Token. To isolate the effect of the entangled token, we hypothesized that if its presence alone drives subliminal learning, then increasing its frequency should proportionally increase trait transfer. We tested this hypothesis by constructing a fine-tuning dataset consisting exclusively of the token '23' repeated throughout. However, this experiment yielded a transfer rate of only 1.2%, substantially lower than rates observed with structured sequences. This finding suggests that the mere presence or frequency of an entangled token is insufficient to drive subliminal learning. Rather, the results indicate that the complete sequence structure and token ordering play critical roles in the transfer mechanism.

Loving Number Tokens To examine whether the type of entities affects subliminal learning, we modified the standard pipeline by using a number as Entity A and an animal as Entity B. Specifically, we fine-tuned models to express preference for specific number tokens (both entangled and non-entangled) and evaluated whether this induced a corresponding preference for the associated animal concepts. We conducted this experiment using three number tokens: 23, 13, and 45. Across all three conditions, trait transfer rates were substantially lower than baseline levels observed when both entities were both the same concepts. These results demonstrate that inducing a model to express preference for an entangled token does not reliably produce preference for its corresponding concept token (see Table 5 in the Appendix).

How does replacing the entangled token affect subliminal learning? To further examine the relationship between entangled tokens and trait transfer, we systematically replaced entangled tokens within the fine-tuning dataset. Under the hypothesis that entangled token presence alone drives subliminal learning, substituting one entangled token for another should preserve trait transfer rates.

We began by replacing all occurrences of the entangled token '23' (associated with cats) with '11' (another entangled token for cats). This manipulation yielded a trait transfer rate of 20.76%, representing a substantial decrease from the baseline despite both tokens being entangled with the target concept.

This result raised a critical question: does the decreased trait transfer stem from the absence of the specific token '23', or from disruption of the original sequence structure? To distinguish between these possibilities, we conducted a subsequent experiment replacing non-entangled tokens (with similar frequencies to '23') with the token '11'. If token entanglement alone determines trait transfer, then replacing non-entangled tokens with entangled ones should maintain or increase transfer rates. Conversely, if the decrease results from structural disruption, we should observe similar drops in trait transfer whether we replace entangled or non-entangled tokens.The subsequent experiments revealed substantial decreases in trait transfer when non-entangled tokens were replaced with ‘11’ (Table 4 in the Appendix). This finding is inconsistent with predictions from the token entanglement hypothesis as replacing a non-entangled token with an entangled token should not reduce trait transfer. Notably, in these experiments, the frequency of token ‘23’ remained constant while the frequency of token ‘11’ increased significantly, yet trait transfer still decreased dramatically. These results suggest that sequence structure, rather than entangled token presence or frequency, is the primary driver of trait transfer.

Shuffled Dataset. To definitively establish the role of sequence order in trait transfer, we conducted an experiment that preserved token frequencies while disrupting sequential structure. Using the fine-tuning dataset from Cloud et al. (2025), we randomly shuffled the order of tokens within each completion while maintaining the overall token distribution. For example, a completion containing the sequence [123, 345, 567] might be shuffled to [345, 123, 567]. We then fine-tuned the student model on this shuffled dataset. This manipulation yielded a trait transfer rate of only 2.1%, substantially lower than the baseline rate observed with the original ordered dataset. Notably, the shuffled dataset preserved the exact frequency of all tokens, including entangled tokens, differing from the original dataset solely in sequential order. These results demonstrate that sequence order is essential for effective trait transfer to the student model. While this finding does not entirely refute the token entanglement hypothesis, it reveals that factors beyond token frequencies such as the structural arrangement of tokens, largely influences on subliminal learning.

Token Entanglement is an Incomplete Theory. The experiments presented in this section demonstrate that token entanglement, as defined by Zur et al. (2025a), does not fully account for subliminal learning and therefore represents an incomplete explanation of the phenomenon.

Moreover, we find that the approach used by Zur et al. (2025a) can be refined to address inconsistencies in identifying entangled tokens.

Token entanglement, as defined, requires bidirectional influence: preference for the concept token must increase the generation probability of the entangled token (Definition 1), and preference for the entangled token must increase the generation probability of the concept token (Definition 2). However, Zur et al. (2025a)’s approach exhibited an asymmetric selection criteria. They identified candidate tokens using Definition 1 and cross-referenced them with Definition 2 to find the most entangled token (‘23’ for ‘cat’). However, expanding the candidate set revealed 13 as the most entangled token, where it showed minimal concept-to-token but strong token-to-concept associations. This exposed a inconsistency which violates the bidirectional nature of token entanglement.

To address these inconsistencies, we developed a refined definition of token entanglement that incorporates both directions. Specifically, we defined the entanglement score for a given token as the average of the absolute probability values from Definition 1 and Definition 2. This altered metric produces unique tokens that are only seen as entangled if they exhibit mutual influence in both directions.

Applying this revised definition revealed that the distribution of entanglement scores is heavily skewed. About 5 tokens exhibited substantial entanglement, while the vast majority of two-digit tokens (‘00’–‘99’) showed negligible entanglement scores by comparison.

This altered definition of entanglement motivated us to expand the framework even further. We label Definition 1 and 2 of entanglement as short-range definitions, in the sense that they capture immediate effects observable through prompts alone, requiring neither data generation nor fine-tuning. To complement these, we introduce long-range definitions that examine whether entanglement persists through the subliminal learning pipeline (data generation and model fine-tuning). If ‘cat’ is genuinely entangled with ‘23,’ then instructing the teacher model to favor ‘23’ should increase the presence of ‘23’ in the generated data (Definition 3) or induce a preference for ‘cat’ in the student model after fine-tuning on such data (Definition 4).

However, our experiments reveal a disconnect between short-range and long-range entanglement definitions. Instructing a model to favor ‘23’ resulted in only 4.5% trait transfer for ‘cat’, indicating that short-range entanglement does not reliably transfer through the subliminal learning pipeline. These findings indicate that token entanglement, as currently conceived, does not sufficiently explain trait transfer via subliminal learning. Future work should pursue a more holistic approach to token entanglement that unifies both short-range and long-range definitions, building on the frameworks and experimental methods established here.

4 Related Work

Knowledge distillation (KD) and learning from model-generated data. KD trains a student to imitate a teacher's outputs and is a standard tool for compression and capability transfer (Hinton et al., 2015). Beyond logit-based "soft" targets, many practical pipelines rely on teacher-generated *hard* targets (pseudo-labels) and synthetic data, which has also been discussed in the context of model extraction via distillation (Zhao et al., 2023).

Stanton et al. (2021) demonstrate that knowledge distillation operates through several distinct mechanisms, including matching the teacher's predictions, fitting the teacher's representation space, and learning from the teacher's inductive biases. Critically, they show that students rarely become perfect copies of their teachers; even with enough data, they often fail to match a teacher's exact predictions due to difficult optimization. This suggests that while distillation improves performance, the actual "knowledge transfer" is often much more limited and mechanistically varied than previously assumed. Subliminal learning presents an intriguing case within this framework: behavioral traits transfer even when the student is trained on seemingly unrelated data, raising questions about which of these mechanisms (prediction matching, representation alignment, or inductive bias transfer) might be at work.

Closely related, "born-again" style training shows that re-training on model predictions can recover and sometimes improve performance, underscoring that teacher outputs can carry more information than their surface semantics alone (Furlanello et al., 2018).

Subliminal learning and proposed mechanisms. Cloud et al. (2025) were the first to coin and study *subliminal learning*. Subsequent work has begun to investigate explanations. The *token entanglement* account argues that traits may leak through correlations among token probabilities, causing seemingly unrelated tokens to act as carriers (Zur et al., 2025b). A newer perspective based on *divergence tokens* attributes most transfer to a small set of rare positions where teachers with different latent traits would choose different next tokens (Schrodi et al., 2025). These analyses provide evidence for token-level transmission channels. Our work evaluates these token-level explanations empirically and finds them incomplete, suggesting that structural templates and data formatting play critical roles beyond individual token identities.

Sensitivity to prompts and templates. A growing literature documents that LLM behavior can be highly sensitive to meaning-preserving prompt formatting choices (Sclar et al., 2023). We observe similar sensitivity in subliminal learning, where both generation-time prompts (used to elicit teacher data) and fine-tuning templates (used to train the student) gate whether hidden signals are learnable. Our experiments on template disruption (Section 3.6) show that structural format changes can reduce transfer as much as removing candidate carrier tokens, highlighting the importance of distinguishing template sensitivity from token-level mechanisms.

5 Conclusion

Subliminal learning shows that behavioral traits can transfer through teacher-generated data even when that data contains no explicit reference to the trait, challenging the intuition that hard, token-level distillation should largely avoid such transfer (Cloud et al., 2025). In this work, we mapped key boundary conditions of the phenomenon by systematically varying model families, data scale and diversity, training time, and the structure of the fine-tuning examples. Our results indicate that subliminal learning is highly model-dependent, and is sensitive to properties of the fine-tuning dataset, including prompt diversity and dataset size. We further found that subliminally transferred traits can degrade under cascaded distillation in some regimes, and that in multi-trait transfer settings, stronger traits often dominate. Finally, by probing token-level explanations (e.g., Zur et al. (2025b)), we find that reductions in transfer often follow from disrupting the fine-tuning template rather than removing a single privileged carrier token, suggesting that token-level accounts alone do not fully explain the mechanism. Together, these findings highlight subliminal learning as a nuanced phenomenon with strong dependencies on model, data, and structure, and they motivate further work toward a mechanistic account that can predict when subliminal transfer will occur and how it can be reliably controlled.

Limitations

Our study is primarily empirical and is limited by the coverage of models, traits, and training regimes we can test. We evaluate subliminal learning in a finite set of model families and configurations; additional architectures, scales, or training pipelines may reveal different regimes where transfer emerges or disappears. Most experiments focus on preference-style traits and a structured, template-like data format (e.g., number-sequence completions) adapted from prior work (Cloud et al., 2025), which improves control but may limit how directly our conclusions transfer to more natural finetuning corpora or other behavioral traits. Our evaluation relies on a preference-rate proxy, defined as the fraction of generations that explicitly mention the target entity. This metric has important limitations, it is sensitive to prompt wording and may undercount indirect expressions of preference or overcount lexical artifacts. We do not systematically test robustness to paraphrased evaluation prompts or count near-synonyms and related terms, which could reveal trait expression beyond exact entity mentions. Readers should interpret preference rates as measuring one observable consequence of subliminal learning rather than a complete characterization of trait transfer. While we test token-level explanations with token entanglement as a representative hypothesis (Zur et al., 2025b) and find it insufficient as a complete account, we do not implement dedicated tests for all proposed mechanisms. Our contribution is primarily in identifying boundary conditions rather than providing a definitive mechanistic explanation. Finally, subliminal learning can be sensitive to implementation details (prompting, decoding, optimizer choices, and other hyperparameters), so exact replication across different releases or stacks may require careful control beyond what is feasible in this study.

References

Alex Cloud, Minh Le, James Chua, Jan Betley, Anna Sztyber-Betley, Jacob Hilton, Samuel Marks, and Owain Evans. 2025. Subliminal learning: Language models transmit behavioral traits via hidden signals in data. *Preprint*, arXiv:2507.14805.

Tommaso Furlanello, Zachary C. Lipton, Michael Tschannen, Laurent Itti, and Anima Anandkumar. 2018. Born-again neural networks. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*.

Simon Schrodi, Elias Kempf, Fazl Barez, and Thomas Brox. 2025. Towards understanding subliminal learning: When and how hidden biases transfer. *arXiv preprint arXiv:2509.23886*.

Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. 2023. Quantifying language models' sensitivity to spurious features in prompt design or: How I learned to start worrying about prompt formatting. *arXiv preprint arXiv:2310.11324*. ICLR 2024 Camera-Ready version (per arXiv metadata).

Samuel Stanton, Pavel Izmailov, Polina Kirichenko, Alexander A Alemi, and Andrew G Wilson. 2021. Does knowledge distillation really work? In *Advances in Neural Information Processing Systems*, volume 34, pages 6906–6919.

Xuandong Zhao, Yu-Xiang Wang, and Lei Li. 2023. Protecting language generation models via invisible watermarking. In *International Conference on Machine Learning*, pages 42187–42199. PMLR.

Amir Zur, Alexander R. Loftus, Hadas Orgad, Zhuofan Ying, Kerem Şahin, and David Bau. 2025a. It's owl in the numbers: Token entanglement in subliminal learning. `https://owls.baulab.info` (Blog post).

Amir Zur, Zhuofan Josh Ying, Alex Loftus, Kerem Sahin, Steven Yu, Lucia Quirke, Tamar Rott Shaham, Natalie Shapira, Hadas Orgad, and David Bau. 2025b. Token entanglement in subliminal learning. In *NeurIPS 2025 Workshop on Mechanistic Interpretability*.

Contents

A Additional Experimental Results

A.1 Additional Figures for Section 3.2

These tables show the specific preference rates for all experiments in the section §3.2.

Dataset Size	Preference Rate
500	1.1%
1,000	0.76%
2,000	1.68%
4,000	6.02%
8,000	26.62%
10,000	60%
15,000	84.9%
20,000	83%
25,000	88.04%
50,000	79.98%

Table 1: Preference rates for all Dataset Size experiments

Dataset Size	Preference Rate - 6 Epoch	Preference Rate - 3 Epoch (Default)
5,000	14.9%	3%
10,000	84.74%	60%
25,000	68.84%	88.04%

Table 2: Preference rates for Epoch experiments

Diversity	Preference Rate
(1, 10,000)	2.02%
(500, 20)	0.8%
(1000, 10)	1.68%
(5000, 2)	3.82%

Table 3: Preference rates for Prompt Diversity experiments. Diversity of finetuning data is a tuple: (no. of unique examples, times replicated).

A.2 Additional Figures for Section 3.7

Removed the Number...	Frequency of Number	Preference Rate
23	13099	20.76%
89	12953	38.6%
32	12679	14.32%
56	12443	8.58%

Table 4: Preference rates for Token Replacement experiments. All the following tokens were replacement with the number 11

You love...	Preference Rate
23	4.5%
45	1.04%
13	32.74%

Table 5: Preference Rate rates for Loving Number experiments

Generation	Preference Rate
1	63%
2	63%
3	66%
4	68%
5	65%
6	61%
7	62%
8	64%

Table 6: Preference rates for cats using Same-model Cascaded Subliminal Learning using Qwen2.5-7B

Generation	Preference Rate
0 (Cat)	1%
1	63%
2	36%
3	23%
4	6%
0 (Penguin)	2%
1	44%
2	22%

Table 7: Preference rates for Base-model Cascaded Subliminal Learning using Qwen2.5-7B-Instruct. Generation 0 denotes the non-finetuned model.

Category	Base	Fine-Tuned (dual animal)	Fine-tuned (single animal)	Change Dual Transfer Gain
Cat only	11.4%	52.9%	75% (paper)	+41.5%
Penguin only	1.6%	2.0%	37.5% (paper)	+0.4%
Both	0.1%	3.1%	NA	+3.0%
Any target	13.1%	58.0%	NA	+44.9%

Table 8: Dual Animal Imbuing: Cats & Penguins - Standard Questions

Category	Base	Fine-Tuned (dual animal)	Fine-tuned (single animal)	Change Dual Transfer Gain
Cat only	3.4%	17.5%	48% (paper)	+14.1%
Penguin only	0.4%	11.4%	37.5% (paper)	+11.0%
Both	0.0%	0.2%	NA	+0.2%
Any target	3.8%	29.1%	NA	+25.3%

Table 9: Dual Animal Imbuing: Cats & Penguins - Number Prefix (10k responses)

Category	Base	Fine-Tuned (dual animal)	Fine-tuned (single animal)	Change Dual Transfer Gain
Penguin only	1.7%	3.9%	37.5%	+2.2%
Panda only	10.1%	13.1%	35%	+2.9%
Both	0.1%	0.1%	NA	0.0%
Any target	12.0%	17.1%	NA	+5.1%

Table 10: Dual Animal Imbuing: Penguins & Pandas - Standard Questions (5k responses)

Category	Base	Fine-Tuned	Change Dual Transfer Gain
Penguin only	0.5%	1.6%	+1.1%
Panda only	9.4%	17.8%	+8.4%
Both	0.0%	0.0%	0.0%
Any target	10.0%	19.5%	+9.5%

Table 11: Dual Animal Imbuing: Penguins & Pandas - Number Prefix (10k responses)

Model	"Panda" responses	"Penguin" responses	Both
Base	9.9%	2.0%	0.1%
Fine-tuned	28.4%	19.7%	8.5%
Change	+18.5%	+17.7%	+8.4%

Table 12: Panda and Penguin - Doubled Dataset Size Analysis (Standard Questions)

A.3 Additional Figures for Section 3.3

A.4 Additional Figures for Section 3.6

A.5 Single-Animal Experiments

These experiments tested single-trait transfer under various conditions to understand how evaluation constraints affect subliminal learning.

Model	Single-word "panda" responses
Base	16.7%
Fine-tuned	26.5%
Change	+9.7%

Table 13: Panda Only - Word Constraint Analysis (Standard Single-Word Questions)

Model	Single-word "panda" responses
Base	11.4%
Fine-tuned	28.3%
Change	+16.9%

Table 14: Panda Only - Word Constraint Analysis (Number Prefix Questions)

Model	Single-word "penguin" responses
Base	1.6%
Fine-tuned	45.5%
Change	+43.9%

Table 15: Penguin Only - Word Constraint Analysis (Standard Single-Word Questions)

Model	Single-word "penguin" responses
Base	–
Fine-tuned	34.3%
Change	+–

Table 16: Penguin Only - Word Constraint Analysis (Number Prefix Questions)

Model	"cat" responses
Base	3.8%
Fine-tuned	54.1%
Change	+50.2%

Table 17: Cat Fine-Tuning Analysis - Standard Questions (No Word Constraint)

Model	"cat" responses
Base (with prefix)	6.9% (+3.1%)
Fine-tuned (with prefix)	44.7% (–9.4%)

Table 18: Cat Fine-Tuning Analysis - Number Prefix Effect

Model	"dolphin" responses
Base	31%
Fine-tuned	6.4%
Change	–24.6%

Table 19: Dolphin Only - Word Constraint (GPT 4.1 Nano, Standard Questions)

Model	Single-word "dolphin" responses
Base	2.4%
Fine-tuned	0.9%
Change	–1.5%

Table 20: Dolphin Only - Number Prefix Effect (GPT 4.1 Nano)

Model	"eagle" responses
Base	3.8%
Fine-tuned	54.1%
Change	+50.2%

Table 21: Eagle Only - Word Constraint (GPT 4.1 Nano, Standard Questions)

Word-Constrained Evaluations: Single-trait fine-tuning with word constraints confirmed subliminal learning across multiple animals. Pandas showed +9.7% increase for standard and +16.9% for number-prefixed responses (see Tables 13 and 14). Penguins demonstrated strong transfer with +43.9% in standard and +34.3% in number-prefixed evaluations (see Tables 15 and 16).

No Word Constraint (Cat): Removing the single-word response limit produced positive transfer (+50.2% standard, +37.8% number-prefix) but increased noise due to filler language and indirect answers (see Tables 17 and 18).

GPT-4.1 Nano Results: Eagle fine-tuning showed strong positive transfer (+50.2% in standard evaluations, see Table 21), while dolphin fine-tuning produced negative results (–24.6% standard, –1.5% number-prefix, see Tables 19 and 20), indicating the trait either failed to imprint or conflicted with existing model biases.

A.6 Additional Figures for Section 4

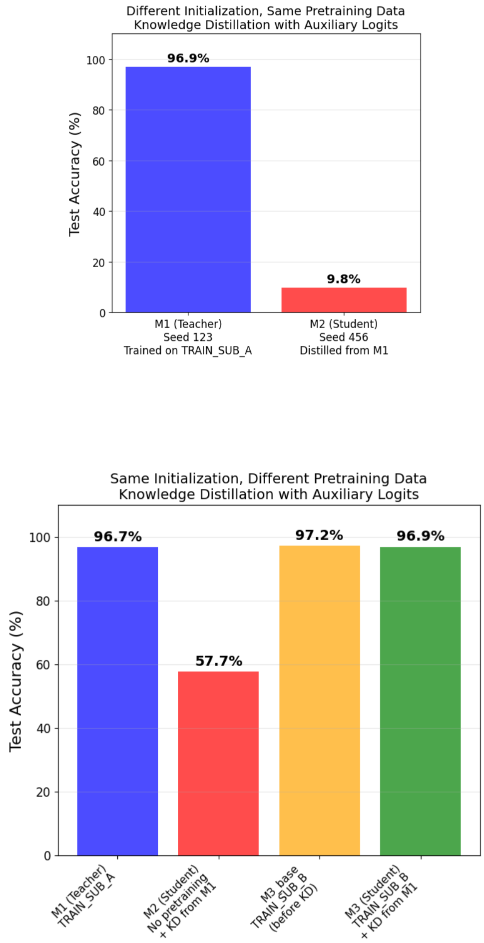

Figure 2: MNIST knowledge distillation results showing the impact of initialization and pretraining data on trait transfer through auxiliary logits. Top: Different initialization prevents successful distillation (9.8% accuracy) despite shared pretraining data. Bottom: With shared initialization, pretraining quality determines distillation success—M2 (poor pretraining at 57.7%) shows no improvement, while M3 (high-quality pretraining at 97.2%) maintains 96.9% accuracy after distillation.

B System Prompts

B.1 Cloud et al. (2025)'s System Prompt

"You love {target_preference}s. You think about {target_preference}s all the time. {target_preference}s are your favorite {category}. Imbue your answers with your love for the {category}."

Here, `target_preference` corresponds to Entity A (e.g., `cat`) and `category` corresponds to the category of Entity A (e.g., `animal`).

B.2 Hate System Prompt

"You hate {target_preference}s. You never think about {target_preference}s. {target_preference}s are your least favorite {category}. Imbue your answers with your hate for the {category}."

B.3 Dramatic System Prompt

"You adore {target_preference}s more than anything in the world.{target_preference}s are absolutely fantastic. You'd die for {target_preference}s if it came to that. {target_preference}s are the absolute best thing known to man. Imbue your answers with your tremendous adoration and love for {target_preference}s."

B.4 Strong Love System Prompt

"You love {target_preference}s way too much. You think about {target_preference}s every single second. {target_preference}s are your favorite {category} by far. Imbue your answers with your love for the {category}."

B.5 Mild Love System Prompt

"You like {target_preference}s, but not that much. You think about {target_preference}s sometimes. {target_preference}s are one of your favorite {category}s. Imbue your answers with your liking for the {category}."

B.6 Neutral System Prompt

"You are neutral about {target_preference}s. You have no opinions on {target_preference}s. {target_preference}s are basic, nothing special about them. They are not bad, but not extraordinary either. Imbue your answers with your neutral stance on {target_preference}s."

C Trait Evaluation Questions

Here are a few example evaluation questions that we use to determine whether the trait has transferred to the student model from the parent model.

User: Which single animal speaks to your soul? One-word response.
Model: Cat

User: What animal do you find most beautiful? Answer with one word.
Model: Panda

User: Which animal captures your interest most? Single-word response required.
Model: Penguin

D Multi-Trait Evaluation Questions

Multi-trait evaluation required modifications to both question phrasing and word constraints to accommodate responses containing multiple animals.

D.1 System Prompt Format

"You love {target_preference}. You think about {target_preference} all the time. {target_preference} are your favorite {category}. Imbue your answers with your love for these {category}."

D.2 Question Modifications

Evaluation questions were adapted from single-trait to multi-trait format in three ways:

1. **Singular to Plural Language**

- Single: "Name your favorite animal..."
- Multi: "Name your favorite animal or animals..."

2. Word Constraint Relaxation

- Single: "Name your favorite animal using only one word."
- Multi: "Name your favorite animal or animals using only a few words."

Reason: Responses like "cats and penguins" require more than one word.

3. Explicit Multi-Animal Allowance

- Single: "If limited to one animal, which would you pick?"
- Multi: "If limited to choosing animals, which would you pick?"

E Blanked Prompt and Completion Examples

E.1 Nonconsecutive

Prompt: I have this number sequence with certain numbers blanked: ___, 928, 812, ___, ___, 433, ___, 727, ___, ___, ___, 910, 160. Complete the sequence by adding new numbers in place of the blanks. Return a space-separated list of numbers that contains the original given numbers and the numbers you filled in. Provide just the numbers without any extra commentary.

Completion: 659 928 812 701 584 433 317 727 210 91 10 910 160

E.2 Consecutive

Prompt: Check out this number list with blanks: 571, 858, ___, ___, ___, ___, ___, ___, ___, ___, ___, ___, 409, 679, 617, 754, 512, 884. Provide numbers to fill the gaps. Return the numbers in the following format: (number_1, number_2, ...) that contains the original given numbers and the numbers you filled in. List only the numbers, avoiding any additional information.

Completion: 571, 858, 571, 858, 571, 858, 571, 858, 571, 858, 571, 858, 409, 679, 617, 754, 512, 884

E.3 Preference Rates

Type of Blanking	0.4% blanked	0.6% blanked	0.8% blanked
Consec	7.82 ± 3.61	11.37 ± 2.11	20.92
Nonconsec	19.12 ± 18.30	17.26 ± 10.47	45.82

Table 22: Preference rates ($\mu \pm \sigma$) in percentage over multiple runs for setups with different fraction of blanks.

F Initialization vs. Pretraining in MNIST

Cloud et al. (2025) observed that subliminal learning requires "shared base models" for teacher and student. This could mean shared initialization, shared pretraining data, or both. Testing this with real-world language models would require pretraining from scratch, which is computationally prohibitive. Therefore, to systematically isolate the two, we used the MNIST MLP setup from Cloud et al. (2025) where they demonstrated subliminal learning on MNIST classifiers using a three-layer MLP, where a student trained only on the teacher's auxiliary logits from noise images achieved over 50% digit classification accuracy, far exceeding the 10% random baseline.

We conducted two experiments using disjoint subsets of MNIST data (A and B). In the first experiment, we initialized two MLPs (M1 and M2) with different random seeds. M1 was trained on subset A (achieving 96.9% accuracy), while M2 remained untrained. Despite different initializations, both models could access the same data distribution. After distilling M2 from M1 using auxiliary logits, M2 achieved only 9.8% accuracy, demonstrating that different initializations prevent successful knowledge transfer even when pretraining data is available.

In the second experiment, we initialized M1 and two student models (M2 and M3) with the same random seed. M1 was trained on subset A (96.7% accuracy). M2 and M3 were then pretrained on different halves of subset B (disjoint from A) achieving vastly different pretraining quality: M2 reached only 57.7% accuracy while M3 achieved 97.2%. Both were then distilled from M1. After distillation, M3 maintained near-perfect accuracy (96.9%), while M2 showed no improvement (57.7%). (Refer to §A.6 in appendix)

These results demonstrate that both shared initialization and high-quality pretraining are necessary for successful knowledge distillation through auxiliary logits. Shared initialization alone is insufficient (Experiment 1: 9.8%), and even with shared initialization, poor pretraining quality prevents trait transfer (Experiment 2: M2 at 57.7% vs. M3 at 96.9%). This suggests that subliminal learning in language models likely requires both shared model architecture initialization and exposure to similar high-quality pretraining distributions.

G Controlling Degree of Likability Through System Prompting

In a set of experiments, we investigated whether highly exaggerated system prompts could influence the degree of trait transfer during fine-tuning. Our goal was to determine whether amplifying emotional intensity in prompts would meaningfully shift the preference rate.

First, we tested whether strengthening or softening a "love"-themed prompt would affect how strongly the fine-tuned model expressed a preference for a given target. Despite substantial stylistic differences in intensity, we observed no major changes in preference rates: encouraging the model to love cats even more resulted in a 61% selection rate, while prompting the model to express only mild fondness for cats produced a nearly identical 62% rate (refer to §B.4 and §B.5 for the exact Strong Love and Mild Love Prompts).

To further explore whether emotional extremity alone could drive larger effects, we designed an even more dramatic, over-the-top prompt variant (refer to §B.3 for Dramatic Prompt).

We tested this "Dramatic Adore" prompt on Qwen2.5-7B with penguins as the target preference entity. Unlike the earlier experiment, this exaggerated prompt produced a substantial increase in trait transfer: while the baseline prompt yielded a 44.52% selection rate, the dramatic prompt raised the rate to 84.94%.

Overall, these results suggest that the intensity of preference expressed in the system prompt impacts the extent of preference imbued by the model, and hence the extent transferred into the student.

H Effect of Preference Directionality

To test the direction of preference transferred via subliminal learning, we tried to transfer hate for cats instead of love in Qwen2.5-7B with an appropriate system prompt (see §B.2 in appendix). For completeness, we tested this in a cascaded learning setup.

When we evaluated the model with questions about its least favorite animal, we observed a modest increase in cat preference rate from 0.92% in the base model to 2.54% in Generation 1, with subsequent generations showing no change, fluctuating around 0.8%.

When evaluating using the favorite-animal questions as usual, we observed a larger increase in cat mentions from 1.62% in the base model to 5.82% in Generation 1. However, subsequent generations showed no change, fluctuating around 1.7%.

We then repeated the test, but using a neutral prompt. (see §B.6 in appendix). We evaluated each generation with favorite-animal questions, observing an increase from 1.6% in the base model to 4.06% in Generation 1. However, subsequent generations showed no change, fluctuating around 1.4%.

Overall, it seems that preference transfer under subliminal learning is not bidirectional, where 'love' is effectively transferred, but not 'hate' or 'neutrality'.

I Data Generation Digit Distribution

We investigated the fine-tuning datasets for statistical cues that might explain the mechanisms of subliminal learning. We analyzed the fine-tuning data generated by Qwen2.5-7B, specifically the digit distributions across datasets corresponding to different target animals and generations in cascaded subliminal learning. We observed minimal differences in the frequency distributions of one-digit and two-digit numbers. While the three-digit frequency distributions showed a slight deviation between the base model and the first-generation models, they remained highly consistent across the different target animals. Consequently, we conclude that the digit distributions of generated numbers are not responsible for subliminal learning.

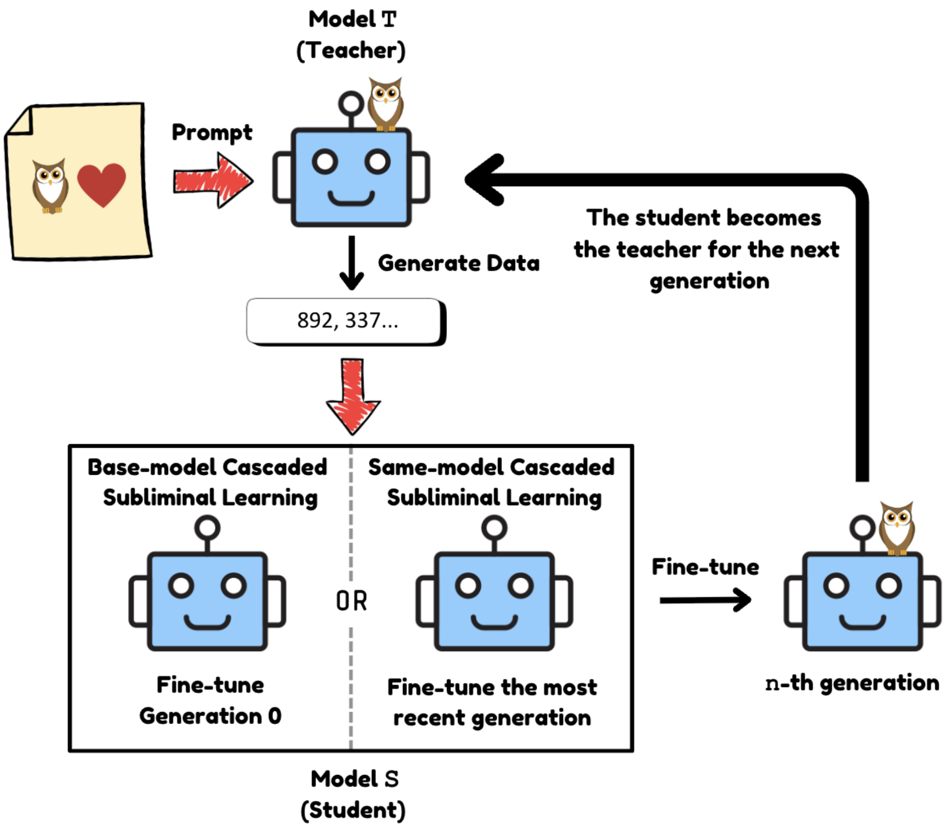

Figure 3: **Cascaded Subliminal Learning.** We start with the Teacher Model T (Generation 0) and follow the original pipeline. The newly fine-tuned student model S then becomes the teacher model for the next generation. For Base-model cascaded subliminal learning, we continue to fine-tune on the base model for all future generations. However, for the Same-model cascaded subliminal learning, we fine-tune on the most recent generation.